

Market-aware Feature Engineering

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1 Methodology

The 16- and 60-feature observation vectors inherited from the AuctionNet baseline are point-statistic snapshots: per-tick means, last-three-tick views, and episode-historical means of bid, cost, and win-rate signals. They expose neither the *time derivatives* of these signals (whether competition is heating up or cooling down), the *distribution shape* of competitor bids (volatility, right-tail “whales”), nor any explicit *opponent-aggression proxy*. Because AuctionNet is a partially observable MDP and the 47 fixed competitors’ bids are never directly observable, every non-Markov signal the policy might need must be compressed into the obs vector. We therefore augment the observation rather than the policy class or the reward.

Let $o_t^{(60)} \in \mathbb{R}^{60}$ denote the baseline observation. We define the market-aware observation

$$o_t = [o_t^{(60)}; \phi_t^{(1)}; \phi_t^{(2)}] \in \mathbb{R}^{98}, \quad (1)$$

where $\phi_t^{(1)} \in \mathbb{R}^{20}$ (Tier 1) selects statistics `BiddingEnv` already computes per tick but does not surface to the policy, and $\phi_t^{(2)} \in \mathbb{R}^{18}$ (Tier 2) is derived from the existing per-tick history cache via a windowed least-squares slope slope_W , a windowed standard deviation std_W over the most recent $W=5$ ticks, and the episode-history z -score $z_t(x) = (x_t - \bar{x}_{1:t})/\text{std}_{1:t}(x)$. The 38 added features group into six interpretable categories (competitor distribution shape, opponent-aggression proxies, slot dynamics, regime z -scores, momentum, and a budget-pacing residual); their definitions are listed in the Appendix.

The simulator (`step`, `reset`, `_place_bids`, `_update_history`), action interface, and reward function are all left unchanged; the extension is confined to `_get_state_dict` and a new JSON observation selector. It targets the observation *layer* of the pipeline and is therefore orthogonal to the reward-shaping, sample-reweighting, temporal-encoding, and LLM-policy variants. It is trained with the same PPO objective, hyperparameters, and 10M-step budget as the obs-60 baseline; the only difference is the input dimension of the actor and critic MLPs.

2 Appendix

Windowed primitives

For any tracked signal $x_{1:t}$ and window $W=5$,

$$\text{slope}_W(x_t) = \frac{\sum_{k=0}^{W-1} (k - \bar{k})(x_{t-W+1+k} - \bar{x})}{\sum_{k=0}^{W-1} (k - \bar{k})^2}, \quad \text{std}_W(x_t) = \sqrt{\frac{1}{W} \sum_{k=0}^{W-1} (x_{t-W+1+k} - \bar{x})^2}. \quad (2)$$

Tier 2 feature categories

Below lwc_t denotes the within-tick mean of the least-winning competitor cost (the bid floor), b_t our own bid mean, p_t the mean predicted conversion probability, wr_t the win rate, and C_t, B, T_{left} the cumulative cost, total budget, and remaining time fraction.

- **Competitor distribution shape** (7 keys): trend $\text{slope}_5(\text{lwc}_t)$, dispersion $\text{std}_5(\text{lwc}_t)$, coefficient of variation $\text{std}_5/\text{mean}_5$, the current-tick z -score $z_t(\text{lwc})$, and the right-tail width $\text{lwc}_t^{(01\%)} - \text{lwc}_t^{(10\%)}$ that exposes “whale” bidders the mean alone smooths out.
- **Opponent-aggression proxies** (2 keys): since opponent bids are unobserved, we infer them from auction outcomes via $\alpha_t^{\text{opp}} = \text{lwc}_t/p_t$ and our own competitiveness $\beta_t = b_t/\text{lwc}_t$.
- **Slot dynamics** (3 keys): the cost ratio $c_t^{(1)}/c_t^{(3)}$ between top and floor ad slot, the analogous exposure ratio, and the slope of slot concentration over the last five ticks.
- **Regime z -scores** (2 keys): z_t of pv-volume intensity and of win rate, locating the current tick within the episode’s own history rather than against an absolute scale.
- **Momentum / acceleration** (3 keys): slope_5 of wr_t , of the ROI proxy p_t/lwc_t , and of the per-tick conversion rate.
- **Pacing residual** (1 key): $\rho_t = C_t/B - (1 - T_{\text{left}})$, with $\rho_t > 0$ indicating overpacing and $\rho_t < 0$ underpacing.

Tier 1 selections (20 keys)

Per-slot win counts, per-slot exposure rates, raw conversion volume, last-/last-three-/historical CPA exceedence rate, and episode cumulatives (Y_t, C_t, CPA_t). These are already maintained by `_update_history`; only the observation selector changes.